

Technical Assessment

Digital Engineering | Engineering Software Developer

Task	Convert a structural engineering spreadsheet into a working software tool
Time allowed	3 to 5 days from receipt of this brief
Deliverable	Working tool and a README, submitted as a ZIP file or repository link
Presentation	60 to 90 minute in-person session with the team

Background

Elastomeric bridge bearings transfer loads between a bridge superstructure and its substructure while accommodating movement and rotation. Before a bearing can be specified on a project, it must pass a series of design checks against the Freyssinet design manual.

Currently, this design check is performed manually using an Excel spreadsheet. Your task is to convert this into a tool that an engineer can run with clearly labelled inputs and outputs.

You do not need to understand the structural theory behind the checks. You do need to translate the spreadsheet logic and produce output that an engineer can verify and trust.

The task

Build a tool that:

1. Accepts the bearing design input parameters listed below.
2. Runs all 7 design checks as defined in the provided spreadsheet.
3. Produces a clearly formatted output showing the result of each check, including key intermediate values and a PASS or FAIL status for each.
4. Displays an overall summary at the end: BEARING PASSES or BEARING FAILS.
5. Validates inputs - it should not crash on bad data; it should tell the user what is wrong.

You are free to choose your own approach, language, and tools. What matters is that the result is correct, clear, and something you can explain and demonstrate in person.

Input parameters

Your tool must accept the following inputs:

Parameter	Description	Units
l, b, T	Bearing plan dimensions and total height	mm
Plate thickness	Thickness of reinforcing steel plates	mm
No. of plates / layers	Number of steel plates and rubber layers	—
te, ti	Edge and internal rubber layer thicknesses	mm
G	Shear modulus of rubber	N/mm ²
Vmax, Vdl, VII	Maximum, dead load, and live load vertical forces	kN
Hs, Ht	Horizontal shear forces, longitudinal and transverse	kN
Long. / Trans. movement	Applied bearing movements	mm
α_b, α_l	Applied rotation angles	radians

The 7 design checks

Your tool must implement all of the following checks. The spreadsheet contains the full calculation logic for each one. Your output must match the spreadsheet results for the sample inputs listed in the Verification section below.

Check	Key intermediate values to show	Pass condition
1. Shear strain	$\delta_b, \delta_l, \delta_r, S, E_q$	$E_q \leq 1.0$
2. Maximum design strain	$A_l, E_c, E_\alpha, k, E_t$	$E_t \leq 7.0$
3. Plate thickness	$t_l, t_2, \sigma_s, t_{min}$	$t_{min} \leq \text{plate thickness}$
4. Stability	$V/A_l, (2 \cdot b_e \cdot G \cdot S') / (3 \sum t_i), b_e/4$	Stress and geometry limits both satisfied

Check	Key intermediate values to show	Pass condition
5. Vertical deflection	E_b , δ , Δ_{Total} , $0.15 \cdot t_i$	$\Delta_{\text{Total}} \leq 0.15 \cdot t_i$
6. Rotational limit	α , $(b_e \cdot \alpha_b + l_e \cdot \alpha_l)/3$	$\alpha \leq (b_e \cdot \alpha_b + l_e \cdot \alpha_l)/3$
7. Fixing of bearings	H , $0.1(V + 2A_l)$, V/A_l	Both sub-checks satisfied

Verification

The spreadsheet includes a worked example. Your tool must reproduce these results exactly using the following sample inputs:

Input	Value	Expected result
Bearing size ($l \times b \times T$)	457 × 254 × 54 mm	Check 1: FAILS ($E_q = 0.80$)
Plate thk / plates / layers	4.5 mm / 4 / 3	Check 2: FAILS ($E_t = 5.43$)
G	0.9 N/mm ²	Check 3: OK
V _{max} / V _{dl} / V _{ll}	1420 / 384 / 1036 kN	Check 4: OK
H _s / H _t	50 / 10 kN	Check 5: FAILS ($\Delta_{\text{Total}} = 1.59$ mm)
Movements (long. / trans.)	10.6 / 4.19 mm	Check 6: OK
α_b / α_l	0.0001 / 0 radians	Check 7: OK
Overall result	BEARING FAILS	

What to submit

- Your complete tool - all source files needed to run it
- A README explaining how to install any dependencies and run the tool, including an example using the sample inputs above
- A brief note on any assumptions you made or anything you would change with more time

Submit as a ZIP file or share a link to a hosted repository.

Rules & guidance

- You may use any language, framework, or tools you prefer - there is no prescribed technology stack
- You may use documentation, reference material, and online resources freely
- The logic for the design checks must live in your code, not in a spreadsheet or external calculation engine
- You will be asked to explain your code in the presentation session, so make sure you understand every part of what you submit
- Please bring your own laptop to present your tool
- If anything in this brief is unclear, get in touch before you start - we would rather answer a question upfront than have you go in the wrong direction

Presentation session

You will have 60 to 90 minutes to present your solution to the team. Plan to spend around 20 minutes walking us through your tool and the choices you made, with the remainder for a live demonstration and questions.

How we will assess your submission

Area	What we are looking for
Correctness	Results match the spreadsheet for all 7 checks using the sample inputs
Code & structure	Solution is readable and organised so that each check is easy to find and follow
Input validation	Handles bad or missing inputs gracefully without crashing
Output quality	Output is clearly labelled, shows intermediate values, and is easy for an engineer to check
Documentation	A colleague could install and run the tool without asking any questions
Presentation	Explains choices clearly, acknowledges limitations, and responds thoughtfully to questions

Good luck - we look forward to seeing what you build.